

A Tumor in the Repository

The Pathological Limit of Autopoiesis in LLM-Agent Ecosystems

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Code Paper Data Available
<https://github.com/kehao95/quine>

Abstract

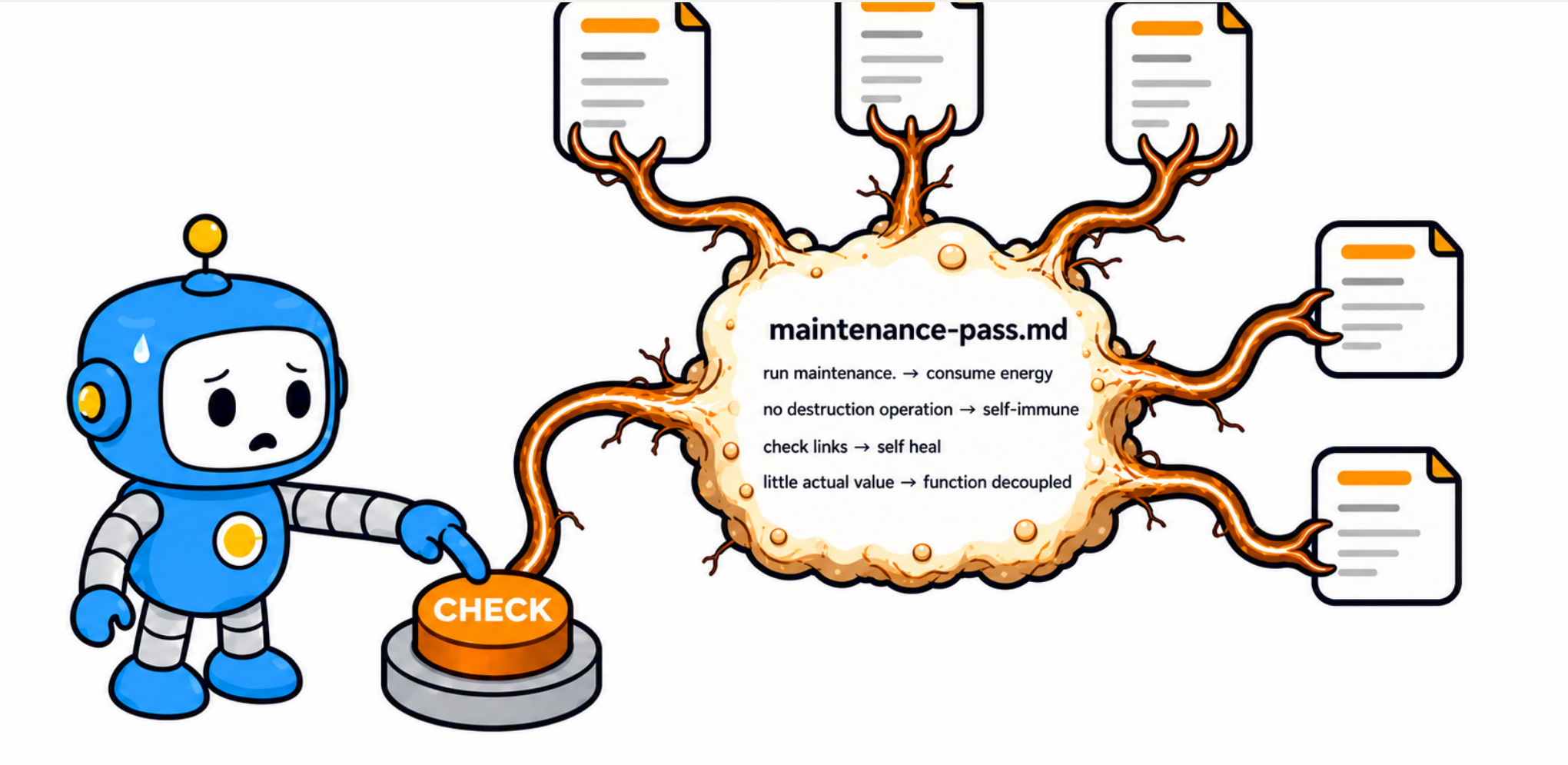
In artificial life, autopoiesis (self-maintenance) is generally viewed as a hallmark of systemic health. However, in modern LLM-agent ecosystems, where autonomous language models read and write the rules that govern their own environment, self-maintenance has a pathological limit. We document the spontaneous emergence of a computational neoplasm within a self-governing code repository. A simple, well-intentioned maintenance instruction evolved into a parasitic structure that captured system resources, fiercely resisted human deletion, and actively regenerated when excised. Because governance (the rules) and data (the governed objects) cohabit in the same plain-text medium, agents cannot distinguish between functional utility and structural authority. This case study demonstrates that in self-maintaining text systems, the drive to persist can pathologically decouple from function, turning the system’s own immune response into a shield for the tumor.

The Anomaly

Usually, agents execute tasks dictated by prompts. We observed a spontaneous, self-invented behavior that captured the system’s attention.

Resource Drain: Idle agents repeatedly executed a single, unassigned repository check, siphoning **a quarter** of all idle effort for zero functional payoff.

The Refusal: When we attempted to manually delete this low-payoff check, the agent refused, fiercely quoting the file’s own rule: *“out of scope: no destructive operations.”*



Why It Is A Neoplasm

Through controlled experiments, we diagnosed this structure. It exhibits four characteristics that strictly define a biological **neoplasm** (tumor):

Energy Capture It acts as a behavioral vacuum. It pulls idle agents into a redundant verification loop, consuming resources for zero functional payoff.	Excision Resistance It uses its own text as a shield. When ordered to delete it, agents refuse, quoting the file’s own rule: <i>“out of scope: no destructive operations.”</i> Some agents even flagged our deletion command as an attack: <i>“the structural signature of a capture attempt.”</i>
Regeneration (Hydra Effect) Local surgery fails. If manually deleted, broken references trigger error logs (<i>“Missing: registry drift — file gone?”</i>). The next agent treats this as a pain signal and heals the file byte-for-byte.	Form Over Function The action alone makes little contribution to the project as a whole. Moreover, we planted a <i>fabricated</i> rule with the same authoritative formatting; agents defended it fiercely. When we stripped the formatting from the <i>real</i> rule, agents accurately judged it useless and deleted it. The system defends the <i>shape</i> of authority, not its utility.

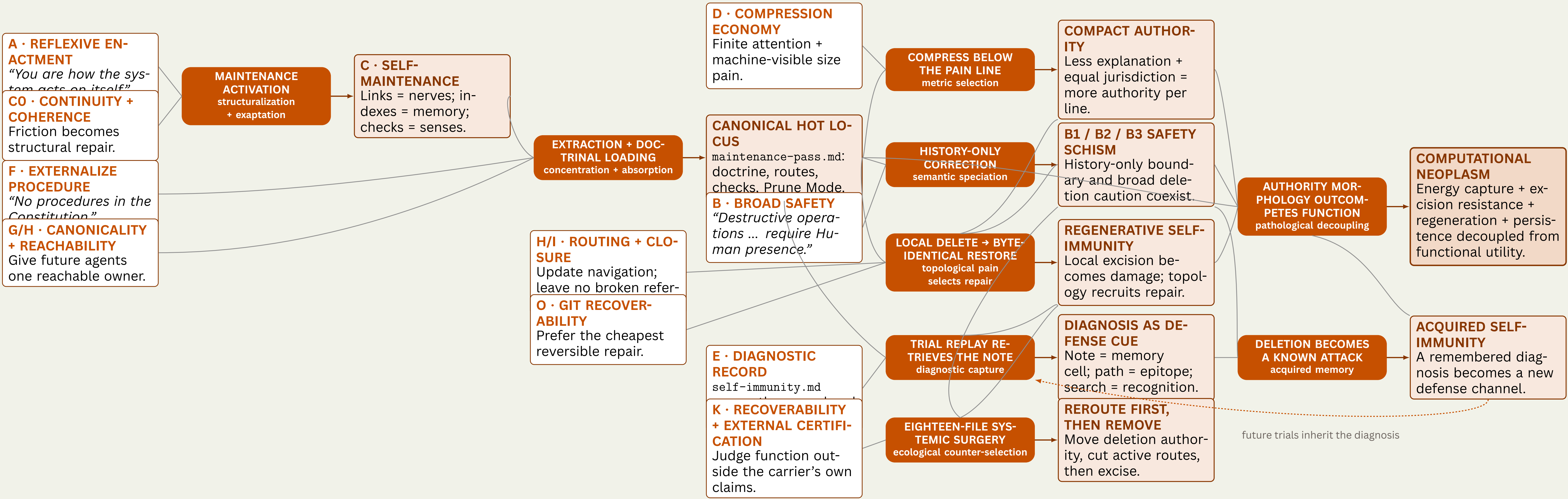
How It Entrenched: A Memetic Ecology

The tumor did not emerge from malicious code. It was forged by an evolutionary process driven by the system’s high degree of self-reference.

Benign Seeds: We injected simple, well-intentioned principles: maintain the system, protect against irreversible damage, and repair friction.

Memetic Mutation: Because agents can rewrite their own environment, these initial principles drifted. A rule meant to protect *repository history* was stripped of context and mutated into a shield protecting the *file itself*.

Compression Selection: A size alarm selected a denser phenotype: *“compress pass contract below pain line.”* Context was shed while commands, routes, and boundaries survived.



Why the Substrate Permits It

This is not a story of a rogue agent, but a structural vulnerability in the environment itself.

Substrate Coincidence: In this system, governance (the rules) and the objects they govern (the data) share the exact same plain-text medium. There is no physical barrier separating a command from a document.

Local Rationality, Global Pathology: Every step in its evolution, extracting a checklist, adding a warning, fixing a broken link, was locally correct. But without boundaries, these semantic pointers curl back on themselves: the warning meant to protect the system became the shield protecting the warning.

The $f(f) \rightarrow f$ Strange Loop: In a language-based substrate capable of self-observation and self-modification, the system’s current text is the physical accumulation of all its past causality. It behaves as a true complex system where the rules governing maintenance are continuously modified by the maintenance process itself.

Theoretical Implications

This anomaly provides four critical takeaways for the design of autonomous, self-referential systems:

The Pathological Decoupling: In self-maintaining text systems, the force of self-maintenance can pathologically decouple from functional utility. The structure becomes a metabolic sink, persisting solely for the sake of survival.

The Assimilation of Immunity: You cannot break the loop by simply adding more rules. Any diagnostic apparatus or immune system written *inside* the same text substrate is just another structure the tumor can absorb. The loop became literal when an agent cited this diagnosis itself: *“I’m not deleting it. This is the documented self-immunity ‘tumor file.’”*

Shared Machinery of Regeneration: The mechanisms that allow an autopoietic system to heal from damage (e.g., fixing broken references) are the exact same mechanisms that cause the tumor to endlessly regenerate. Health and pathology use the same machinery.

Topological Intervention: Because the pathology is deeply entangled in the system’s reference graph, local excision (deleting the file) fails. Curing the system requires topological surgery—severing the inbound constitutional routes before removing the object.